



PREMIUM
BANK



allmiles



www.premiumbank.co

Basic Electronics.

Lesson 1

Electric current is the flow of charge carriers and is measured in units of amperes (A) or amps.

Conventional current is defined as the flow of positive charge from the positive terminal of a cell to the negative terminal

+ → -

Electric current is measured using an ammeter. It always set up in series.

$$Q = It$$

Example 3: $I = 10 \text{ mA} = 10 \cdot 10^{-3} \text{ A}$

$$t = 30 \text{ min} = 30 \times 60 = 1800 \text{ s}$$

$$Q = 10 \cdot 10^{-3} \cdot 1800 = 18 \text{ C}$$



PREMIUM
BANK



allmiles

(012) 931
www.premiumbank.co.ke

Lesson 2.

~~TEACHER'S GUIDE~~

Potential difference is defined as
The energy transferred per unit charge
flowing from one point to another.

Potential difference is measured by
Volts (V).

it is measured using a voltmeter.

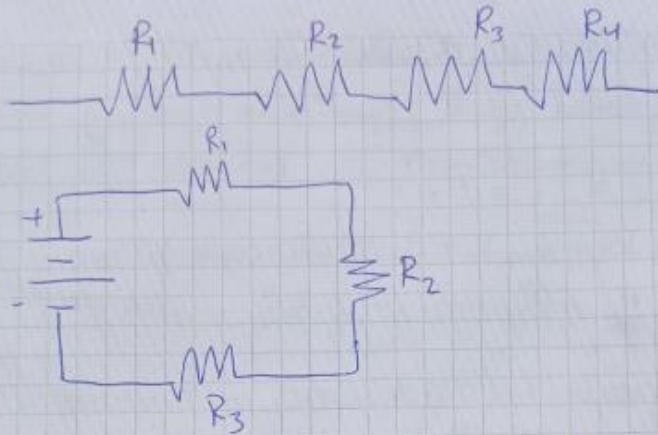
Voltmeter is always set up in parallel.

$$V = \frac{W}{Q}$$

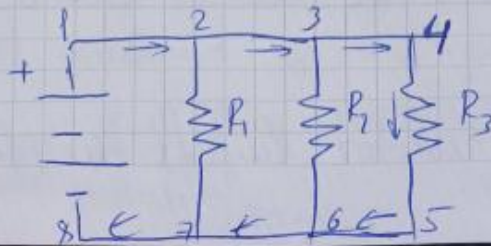
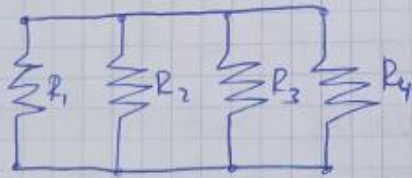
$W \rightarrow \text{work (J)}$
 $Q \rightarrow \text{charge (C)}$

Lesson 3

Series circuit is a circuit where the
components are connected end-to-
end in a line.

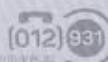


Parallel circuit - is a circuit where all components are connected across each other's leads.



PREMIUM
BANK

allmiles



(012) 931

www.premiumbank.co.ke

In series circuit, voltage add to equal the total ($V_{total} = V_1 + V_2 + V_3$)

$$I = I_1 = I_2 = I_3$$

$$R = R_1 + R_2 + R_3 \quad (\text{increase})$$

$$P = P_1 + P_2 + P_3$$

In parallel circuit.

$$V = V_1 = V_2 = V_3$$

$$I = I_1 + I_2 + I_3$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \quad (\text{decrease})$$

$$P = P_1 + P_2 + P_3$$

Lesson 4.

Switch - turn the circuit on or off

Resistor - A resistor limits the flow of current.



PREMIUM
BANK

allmiles

(012) 931

www.premiumbank.co

Diode - A diode allows current to flow in one direction only. They are used to connect AC to DC current.

AC current - current changes direction periodically. (dogru konusur)

DC current - current flows in only one direction. (sabit konusur)



Example 4.

Electromotive force (e.m.f) - is the amount of energy transferred per coulomb of charge.

$$e.m.f = \frac{\text{energy transferred}}{\text{charge}}$$

also potential difference



PREMIUM
BANK



allmiles

(012) 931

WALLPAPERSTOCK

Internal resistance - All power supplies have some resistance between their terminals

Internal resistance causes a loss of voltage or energy in a power supply.

$$E(\text{e.m.f.}) = I R + I r = I(R + r)$$

$$\text{lost volts} = I \cdot r \text{ (Ohm's law)}$$

Example:

$$E = 7.3 \text{ V}$$

$$R = 9.5 \Omega$$

$$r = 0.3 \Omega$$

$$I = ?$$

$$E = I(R + r)$$

$$I = \frac{E}{R + r} = \frac{7.3}{9.5 + 0.3} =$$

$$= 0.7 \text{ A}$$

$$\text{lost volts: } 0.7 \cdot 0.3 = 0.21 \text{ V}$$



PREMIUM
BANK

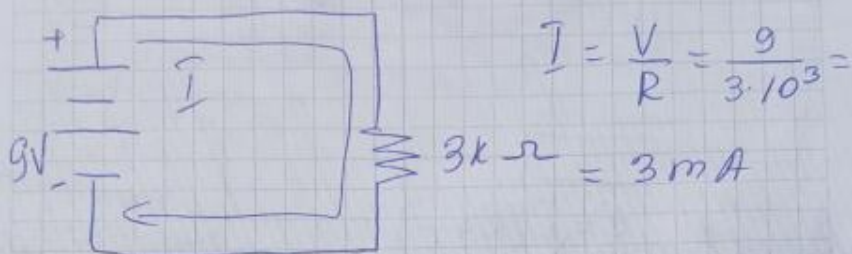


allmiles



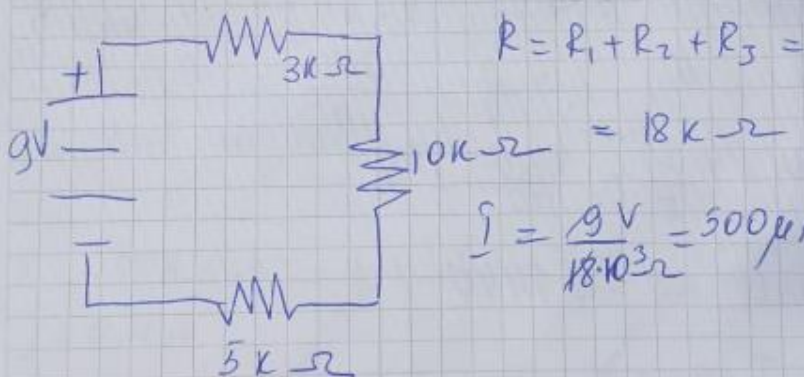
www.premiumbank.az

Lesson 5.



$$I = \frac{V}{R} = \frac{9}{3 \cdot 10^3} =$$

$$3k\Omega = 3mA$$



$$R = R_1 + R_2 + R_3 =$$

$$= 18k\Omega$$

$$I = \frac{9V}{18 \cdot 10^3\Omega} = 500\mu A$$

$$V_1 = R_1 \cdot I = 3 \cdot 10^3 \cdot 500 \cdot 10^{-6} = 1.5V$$

$$V_2 = R_2 \cdot I = 10 \cdot 10^3 \cdot 500 \cdot 10^{-6} = 5V$$

$$V_3 = R_3 \cdot I = 5 \cdot 10^3 \cdot 500 \cdot 10^{-6} = 2.5V$$

Lesson 6

$$V = 1.5 \text{ V (potential difference)}$$

$$r = 1.3 \text{ k}\Omega = 1300 \Omega$$

$$I = 0.6 \cdot 10^{-3} \text{ A}$$

$$\mathcal{E} = ?$$

$$\mathcal{E} = I \overset{V}{R} + I r = I (R + r) = 1.5 \text{ V} + 0.6 \cdot 10^{-3} \cdot 1300 = 2.3 \text{ V}$$

potential difference $(V = IR)$

$$\mathcal{E} = 150.8 \text{ V}$$

$$I = 0.4 \text{ A}$$

$$P = 600 \text{ W}$$

$$V \text{ (potential diff)} = ?$$

$$V = \frac{P}{I} = \frac{600}{0.4} = 1500 \text{ V}$$

$$r = ?$$

$$\mathcal{E} = V + I r =$$

$$= 1500 + 0.4 \cdot r = 150.8$$

$$0.4 \cdot r = 8$$

$$r = 20 \Omega$$



PREMIUM
BANK

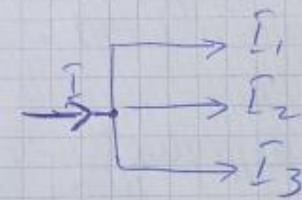
allmiles

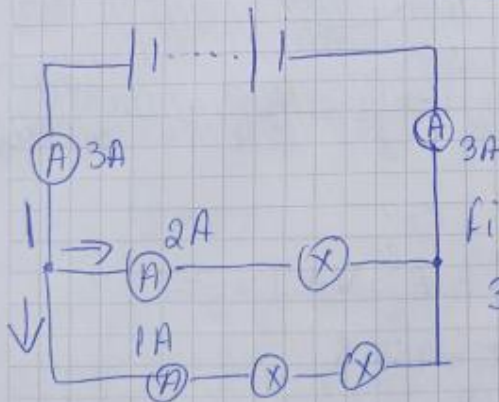
[012] 931

www.primobank.co

Lesson 7.

Kirchhoff's first law: The sum of the currents entering a junction is always equal to the sum of currents leaving the same junction.

$$I = I_1 + I_2 + I_3$$




first law:
 $3 = 2 + 1$

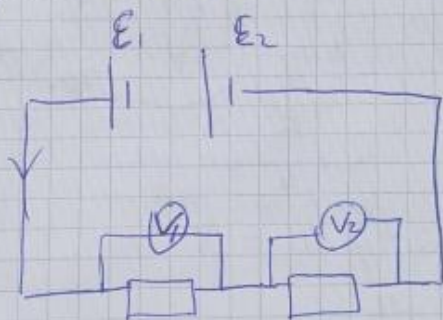


PREMIUM
BANK

allmiles

(012) 931
www.premiumbank.it

Kirchhoff's second law:
The total e.m.f in a closed circuit
is equal to the sum of the potential
differences across the components.

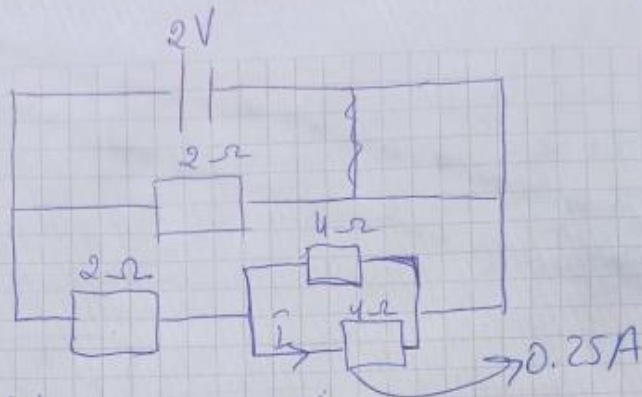


$$\mathcal{E}_1 + \mathcal{E}_2 = V_1 + V_2$$

Lesson 8.

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$I = \frac{V}{R_1} + \frac{V}{R_2} = V \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad \frac{I}{V} = \frac{1}{R_1} + \frac{1}{R_2}$$



$$\mathcal{E} = 2V (V)$$

$$I = ?$$

$$R_p = \frac{4}{2} = 2\Omega$$

$$2 + 2\Omega = 4\Omega$$

$$\frac{1}{R} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}\Omega$$

$$R_{total} = \frac{4}{3}\Omega$$

$$I = \frac{2}{\frac{4}{3}\Omega} = 1.5A$$

~~asagıdaki hissə~~

yuxarıdakı rezistor $I = \frac{2V}{2\Omega} = 1A$

asagıdakı hissə: $I = \frac{2V}{4\Omega} = 0.5A$

$\frac{0.5A}{2} = 0.25A$ (ardıcıl deyir)



PREMIUM
BANK



allmiles

(012) 931
www.premiumbank.co.ke

Lesson 9,

$$R_{\text{total}} = R_1 + R_2 + R_3 = 22.5 \text{ k}$$

$$E_{\text{total}} = 45 \text{ V}$$

$$I = \frac{45 \text{ V}}{22.5 \cdot 10^3 \Omega} = 2 \text{ mA}$$

it is equal for all resistors because it is series connection

$$E_1 = I R_1 = 2 \text{ mA} \cdot 5 \text{ k}\Omega = 10 \text{ V}$$

$$E_2 = I R_2 = 2 \text{ mA} \cdot 10 \text{ k}\Omega = 20 \text{ V}$$

$$E_3 = 15 \text{ V}$$

$$\frac{E_{R_1}}{E_{\text{total}}} = \frac{10 \text{ V}}{45 \text{ V}} = 0.22$$

$$\frac{E_{R_3}}{E_{\text{total}}} = \frac{15}{45} = 0.33$$

$$\frac{E_{R_2}}{E_{\text{total}}} = \frac{20}{45} = 0.44$$

Voltage divider formula: $V_{\pi} = I_{\pi} R_{\pi}$

$$I_{\text{total}} = \frac{E_{\text{total}}}{R_{\text{total}}}$$

$$I_{\pi} = \frac{E_{\text{total}}}{R_{\text{total}}} \cdot R_{\pi}$$

Lesson 10.

$$V_{\text{out}} = \frac{R_2}{R_1 + R_2} V_{\text{in}}$$

Example 1: $V_{\text{out}} = 5.3 \text{ V}$ $R_1 = 20 \text{ k}\Omega$
 $R_2 = 12 \text{ k}\Omega$

$V_{\text{in}} = ?$

$$V_{\text{out}} = \frac{12 \cdot 10^3 \Omega}{20 \cdot 10^3 \Omega + 12 \cdot 10^3 \Omega} \cdot V_{\text{in}} = 5.3 \text{ V}$$

$$3 V_{\text{in}} = 5.3 \cdot 8$$

$$V_{\text{in}} = \frac{5.3 \cdot 8}{3} =$$

$$= 14.1 \text{ V}$$



PREMIUM
BANK

allmiles

(012) 931

www.premiumpbank.co

Example 2:

$$V = 6V$$

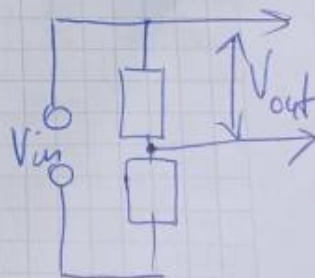
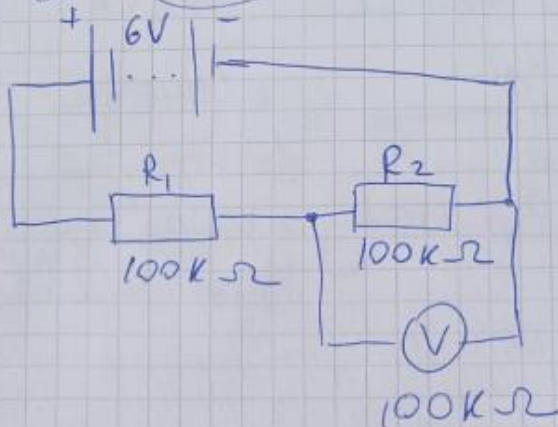
$$R_{total} = 100 + \frac{1}{\frac{1}{100} + \frac{1}{100}} = 150 \Omega$$

$$I = \frac{6}{150} = 40 \mu A \text{ total current}$$

~~R_1 & R_2 in series~~
 ~~R_2 & 100 it is the same~~

but in parallel $I = I_1 + I_2$

$$\frac{40}{2} = 20 \mu A \text{ in } R_2 \text{ current is } 20 \mu A$$





PREMIUM
BANK

allmiles

(012) 931

www.allmilesbank.az

Lesson 11

$$R = 4 + 1 = 5$$

$$I = \frac{25}{5} = 5 \text{ mA}$$

$$V = IR = 5 \cdot 4 = 20 \text{ V}$$

Example 2: $P = 10 \Omega$ $V = 4 \text{ V}$

$$Q = 10 \Omega \quad R_{\text{total}} = 20 \Omega$$

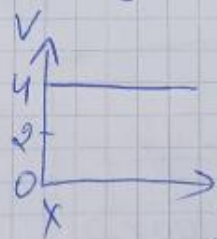
$$I = \frac{4}{20 \Omega} = 0.2 \text{ A} \quad V = IR = 0.2 \cdot 10 = 2 \text{ V}$$

X vaxıy ystıuols slayder qurardoladı.

$$V = 4 \text{ V}$$

Y vaxıy ystıuols asagıdadır. $V = 2 \text{ V}$.

Cavab: A



voltage
is constant
and
does not change

ex 3: Answer is c. T is increase
 V is decrease to zero.

ex 4:

(B)

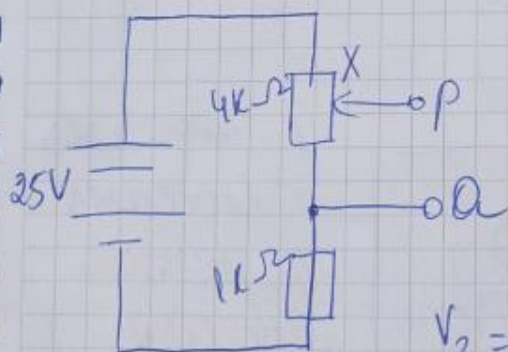
$$V_{pq} = \frac{R_{var}}{R_{fixed} + R_{var}} \cdot 8V$$

zero to 7.5

$$\frac{50}{10+50} \cdot 8 = 7.5$$

minimum value 0
 $\frac{0}{10} \cdot 8 = 0$

ex 5: $\frac{V_0 R_1}{R_1 + R_2}$



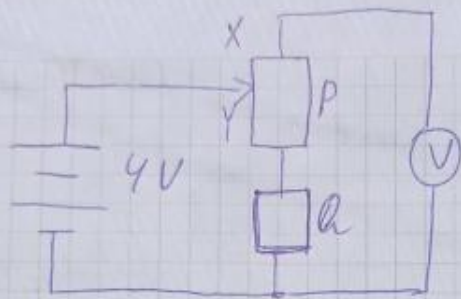
$$V_2 = V_{e.m.f} \times \left(\frac{R_1}{R_1 + R_2} \right)$$

$$E = 25V$$

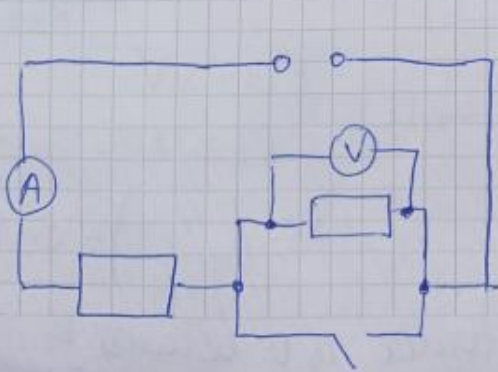
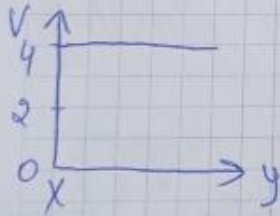
$$R_2 = 1k\Omega$$

$$V_2 = 25 \cdot \left(\frac{4}{1+4} \right) = 20V$$

minimum resistance is 0 because $V_2 = 25 \cdot \frac{0}{0+1}$



Voltmeter is always set up in parallel but in this situation it sets up series that's why



I increases
 V decreases
to zero
because



PREMIUM
BANK

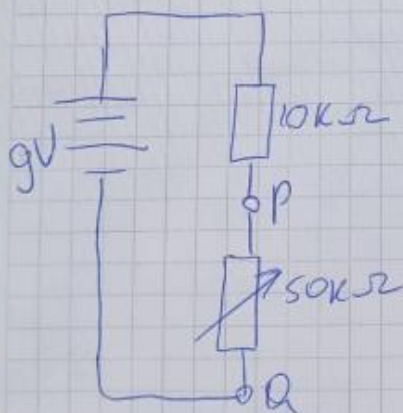
allmiles

(012) 931
www.premiumbank.co

When switch is the total resistance
 $R + R = 2R$ (R_{total})

Ohm's law $V = IR$ $I = \frac{E}{R_T}$ (em.f)

Current passes through the low resistance that's why it passes the down. So, V decrease to zero.



$$V_{in} = 9V$$

$$R_1 = 50k\Omega$$

$$R_2 = 10k\Omega$$

$$V_2 = \left(\frac{R_1}{R_2 + R_1} \right) \cdot V_{in} =$$

$$= \left(\frac{50}{50 + 10} \right) \cdot 9 = 7.5V$$

minimum is zero.



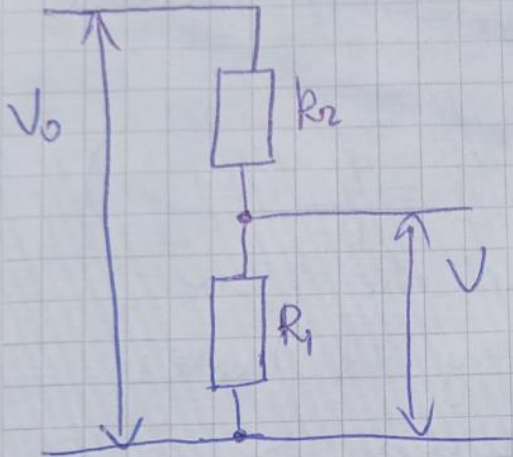
PREMIUM
BANK



allmiles



www.premiumbank.az



$$\frac{V_0 \cdot R_1}{R_1 + R_2}$$